Predictive Maintenance and Explainable Machine Learning

Detecting industrial equipment failures under severe class imbalance: a comparison of three model families and interpretation through Shapley values

> *"To what extent can Machine Learning models effectively detect failure risk, and which factors most influence their predictions?"*

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DATASET	**AI4I 2020 Predictive Maintenance Dataset** — UCI Machine Learning Repository
METHODS	Logistic Regression · Random Forest · XGBoost · SHAP
ENVIRONMENT	Python · scikit-learn · XGBoost · SHAP · Pandas
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1. Abstract

This study evaluates the ability of Machine Learning models to anticipate industrial equipment failures from operating variables, and identifies the factors driving their predictions.

Three model families — logistic regression, random forest and gradient boosting — are compared on the AI4I 2020 dataset (10,000 observations, 3.39% failures) under an identical protocol: stratified splitting, class weighting, five-fold cross-validation, and evaluation by PR-AUC, precision and recall.

XGBoost achieves the best performance (PR-AUC = 0.813, against 0.774 for random forest and 0.382 for logistic regression). Shapley value analysis attributes 75% of total importance to the three mechanical variables — torque, tool wear and rotational speed — and reveals non-linear, threshold-driven relationships.

The initial hypothesis, that non-linear models would produce more false positives, is **refuted**: logistic regression generates nine times as many as XGBoost. This result is explained by the geometry of the problem, failures occurring in two opposing mechanical regimes that no linear boundary can isolate simultaneously.

2. Problem statement

2.1 Industrial context

Unplanned equipment failures halt production lines and generate two distinct cost categories: lost output during downtime, and the premium of an emergency intervention over scheduled maintenance.

Anticipating these failures would allow interventions to be planned. The difficulty lies in a timing trade-off: intervening too early means replacing components that are still functional and taking the machine offline unnecessarily; intervening too late allows the very failure one sought to prevent. This trade-off structures the entire study, from the choice of evaluation metrics to the setting of the decision threshold.

2.2 Research question and hypothesis

> ***Research question.*** *To what extent can Machine Learning models effectively detect failure risk, and which factors most influence their predictions?*

> ***Hypothesis.*** *Non-linear models should detect failures more effectively than logistic regression, but may produce more false positives.*

3. Data

3.1 Description

The AI4I 2020 dataset, from the UCI Machine Learning Repository, comprises 10,000 observations described by 14 variables. It contains neither missing values nor duplicates. The target variable `Machine failure` records **339 failures, or 3.39% of observations**.

The data is **synthetic**: it reproduces realistic industrial behaviour through deterministic generation rules, rather than originating from physical sensors. This characteristic is addressed in the discussion of limitations.

3.2 Variable selection and data leakage

Of the thirteen variables available besides the target, six are retained. The exclusions rest on two distinct justifications.

Identifiers. `UDI` and `Product ID` each hold 10,000 distinct values across 10,000 observations. Neither carries generalisable information.

Failure mode indicators. The variables `TWF`, `HDF`, `PWF`, `OSF` and `RNF` designate the *type* of failure observed. They are recorded only once the failure has occurred and would therefore be unavailable at the moment a prediction must be produced. Retaining them as explanatory variables would constitute **data leakage**: the model would display near-perfect test performance while being inoperative in production.

Variable	Role	Nature
`Type`	Explanatory	Categorical (L / M / H)
`Air temperature [K]`	Explanatory	Continuous
`Process temperature [K]`	Explanatory	Continuous
`Rotational speed [rpm]`	Explanatory	Continuous
`Torque [Nm]`	Explanatory	Continuous
`Tool wear [min]`	Explanatory	Continuous
`Machine failure`	Target	Binary
`UDI`, `Product ID`	Excluded	Identifiers
`TWF`, `HDF`, `PWF`, `OSF`, `RNF`	Excluded	Data leakage

3.3 Labelling anomalies

The count of failure modes (373 occurrences) exceeds that of recorded failures (339). Three phenomena explain this: 24 observations carry several simultaneous modes; 18 observations show a mode raised without a failure label, all of them `RNF` (*Random Failure*), a stochastic failure mode unpredictable by construction; and 9 failures are declared without any identified mode, which constitutes a labelling defect.

These observations are retained. Removing them would amount to cleaning the test set using information unavailable in production. The nine failures without identifiable cause nonetheless lower the attainable performance ceiling.

3.4 Signal structure

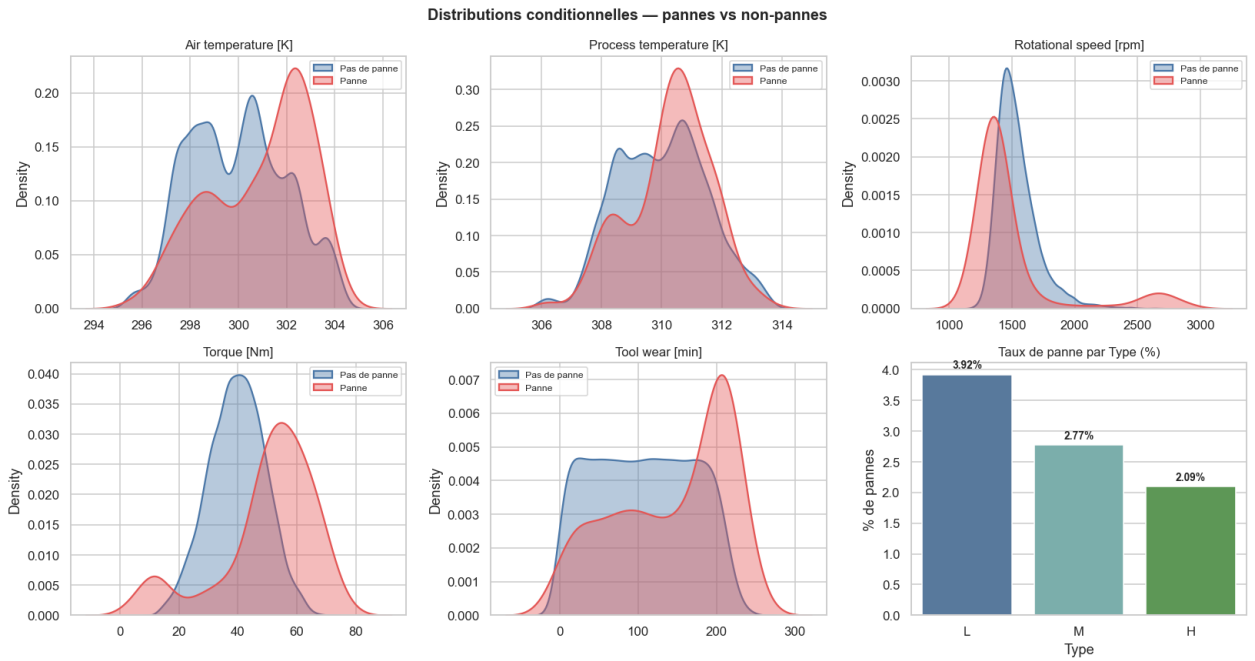


Figure 1. Distributions of operating variables by failure occurrence. Torque and tool wear separate the two classes clearly; the temperatures overlap substantially.

Three variables visibly discriminate between classes. Torque is on average 27% higher under failure, with a **bimodal conditional distribution**: a principal mass between 40 and 70 Nm and a second group below 15 Nm. Rotational speed shows a mirrored profile, consistent with the mechanical relationship linking these two quantities. Tool wear is on average 35% higher before a failure.

Linear correlations with the target nonetheless remain weak (0.191 at most, for torque), suggesting that failures arise from **combinations of conditions** rather than from a threshold being crossed on any single variable.

Two strong collinearities are noted: between the two temperatures ($r \approx 0.88$) and between torque and rotational speed ($r \approx -0.88$). Both affect the interpretation of logistic regression coefficients and the distribution of importance across redundant variables.

4. Methodology

4.1 Experimental protocol

1. **Stratified train/test split** (80 / 20), stratification preserving the failure rate in both samples.
2. **Encoding** of the categorical variable `Type` as indicator variables, with `H` as the reference level.
3. **Standardisation** applied to logistic regression only, via a pipeline. Tree-based models operate through univariate thresholds and are invariant to monotonic transformations; they are trained on variables in their original units, which preserves the readability of the interpretation plots.
4. **Imbalance compensation** through weighting inversely proportional to class frequency, applied identically to all three models to ensure a fair comparison.
5. **Stratified cross-validation** over five folds on the training sample.
6. **Evaluation** on the held-out test sample.
7. **Decision threshold tuning** from out-of-fold predictions, the test set never being consulted for this adjustment.
8. **Interpretation** of the best model through Shapley values, followed by error analysis.

4.2 Choice of metrics

With 3.39% failures, a classifier predicting the systematic absence of failure reaches **96.6% accuracy** without detecting a single breakdown. Accuracy is therefore discarded.

PR-AUC is adopted as the primary metric. Unlike ROC-AUC, it does not benefit mechanically from the abundance of true negatives and faithfully reflects the difficulty of the imbalanced problem. Being threshold-independent, it measures intrinsic discriminative power.

Precision and **recall** are examined jointly: they translate the business trade-off directly, recall measuring the share of failures detected and precision the share of alerts that are justified.

4.3 Rationale for the imbalance treatment

With 28.5 negative observations per positive one, a model optimising an unweighted loss has every incentive to ignore the minority class. Class weighting is preferred over SMOTE-style resampling, which would introduce synthetic observations into an already simulated dataset.

5. Results

5.1 Cross-validation

Model	Mean PR-AUC	Standard deviation
Logistic Regression	0.441	± 0.030
Random Forest	0.750	± 0.037
XGBoost	0.751	± 0.036

The gap between logistic regression and the two ensemble models far exceeds the dispersion across folds: it is robust. The gap between random forest and XGBoost (0.001), by contrast, is far smaller than their respective standard deviations: **these two models are statistically indistinguishable** at this stage.

5.2 Test set performance

The test sample comprises 2,000 observations, of which 68 are failures.

Model	Precision	Recall	F1	PR-AUC	ROC-AUC
Logistic Regression	0.142	0.824	0.242	0.382	0.907
Random Forest	0.808	0.618	0.700	0.774	0.971
XGBoost	0.609	0.824	0.700	**0.813**	0.972
XGBoost — tuned threshold	0.828	0.706	0.762	0.813	0.972

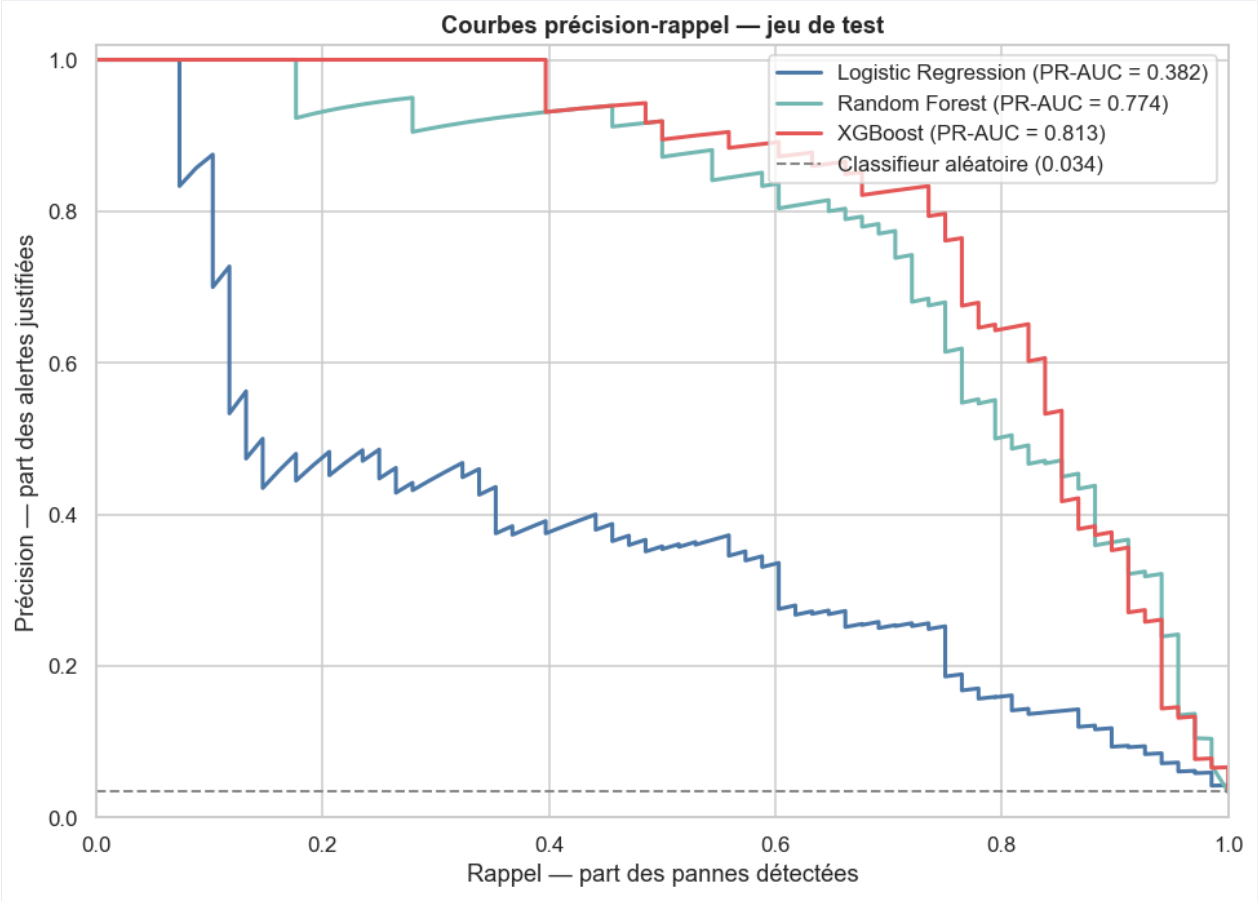


Figure 2. Precision-recall curves on the test set. The horizontal line marks the level achieved by a random classifier (0.034).

Logistic regression fails on precision, not on recall. Its recall (0.824) matches that of XGBoost: it detects as many failures as the best model. It is its precision that collapses to 0.142 — fewer than one alert in seven is justified.

Random forest and XGBoost reach an identical F1 (0.700) through opposing trade-offs. The former favours precision (0.808) at the expense of recall (0.618); the latter does the reverse. This coincidence of scores illustrates the limitation of an aggregate metric considered in isolation.

5.3 Operational implications

Model	Failures detected	Failures missed	False alerts	Interventions triggered
Logistic Regression	56 / 68	12	339	395
Random Forest	42 / 68	26	10	52
XGBoost	56 / 68	12	36	92
XGBoost — tuned threshold	48 / 68	20	10	58

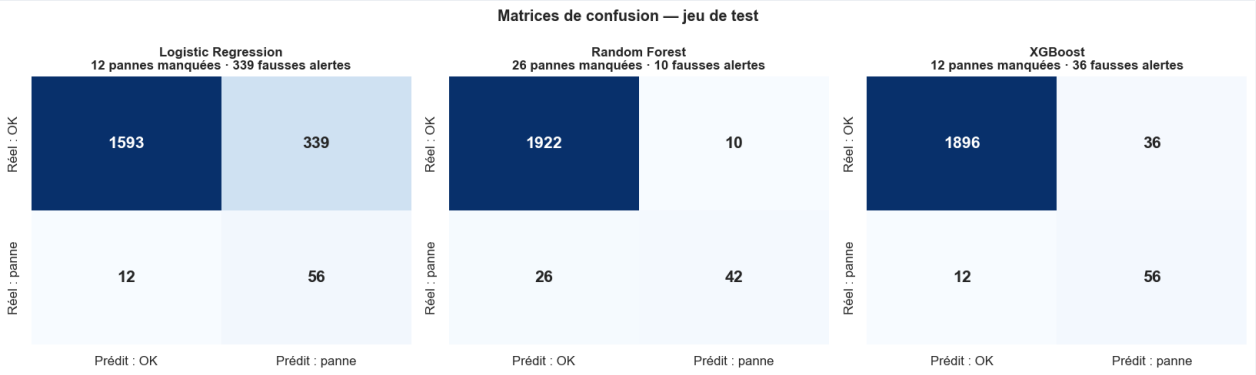


Figure 3. Confusion matrices for the three models on the test set, at a decision threshold of 0.50.

In operational terms, logistic regression would trigger **395 interventions for 56 actual failures** — close to seven maintenance call-outs per failure averted, a system that would be abandoned within weeks. Random forest exhibits the symmetric flaw: only 10 false alerts, but more than one failure in three left undetected.

5.4 ROC-AUC versus PR-AUC

All three models display a high ROC-AUC, reaching 0.907 for logistic regression — which nonetheless issues only one justified alert in seven. ROC-AUC incorporates the true negative rate, abundant here, and produces a flattering picture of an unusable model. The PR-AUC of the same model falls to 0.382 and reflects the situation correctly.

This contrast constitutes the **empirical justification for selecting PR-AUC** as the primary metric under severe class imbalance.

6. Testing the hypothesis

> **PRINCIPAL FINDING**
>
> The hypothesis is **confirmed on detection and refuted on false positives**.
>
> Non-linear models do discriminate considerably better: PR-AUC rises from 0.382 to 0.813. But far from producing *more* false positives, XGBoost generates **nine times fewer** than logistic regression — 36 against 339.

The explanation lies in the **geometry of the problem** rather than in the intrinsic capacity of the models.

Exploratory analysis established that failures occur in two opposing mechanical regimes: overload (high torque, low speed) and underload (low torque, high speed). These two regions occupy opposite extremes of the same variable.

Logistic regression can separate classes only by a hyperplane, that is, through a monotonic effect on each variable. No such boundary can isolate two regions at opposite ends of an axis without enclosing the intermediate space — precisely where the bulk of healthy observations lie. To capture both regimes, the model is forced to widen its alert zone dramatically, which produces the 339 false positives observed.

Tree-based models, by recursively partitioning the variable space, isolate each regime separately without including the central zone.

It is therefore not the “power” of the model that explains the gap, but **the fit between the shape of its decision boundary and the geometric structure of the problem**.

7. Model interpretation

7.1 Variable hierarchy

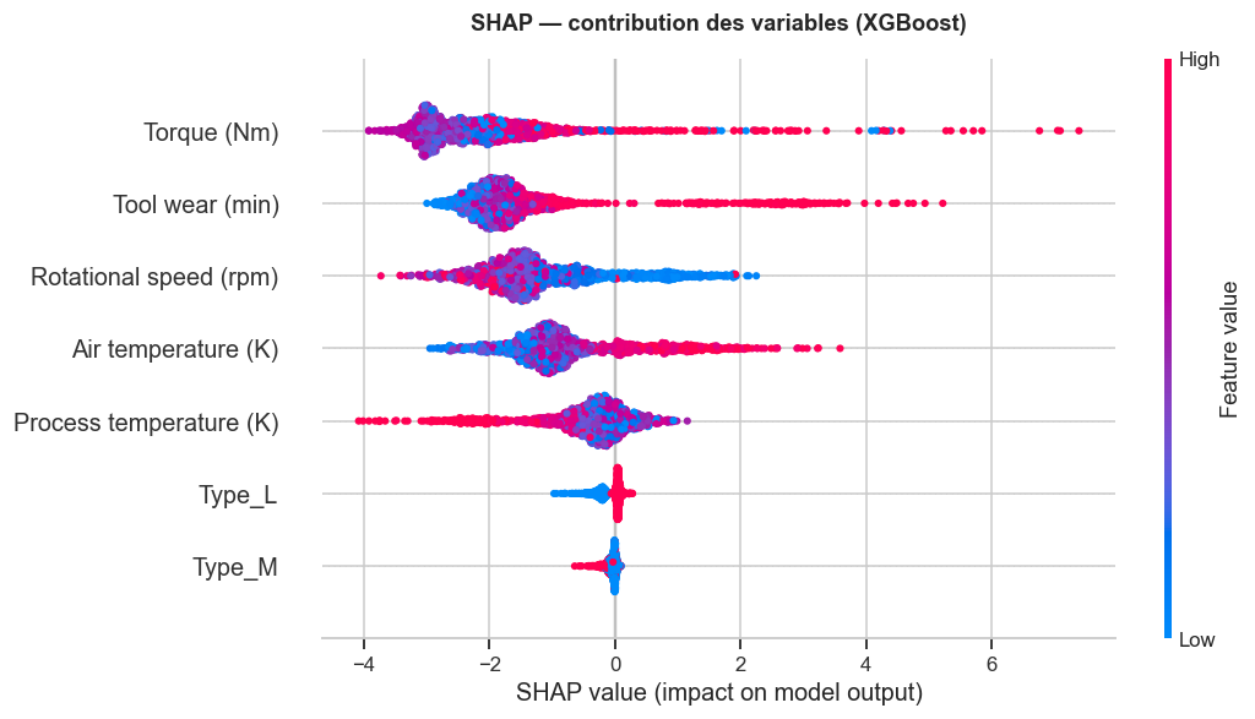


Figure 4. Variable contributions to XGBoost predictions (Shapley values). Each point is an observation; colour encodes the variable's value, the horizontal axis its contribution to the failure score.

Variable	Share of total importance
Torque (Nm)	30.8%
Tool wear (min)	25.1%
Rotational speed (rpm)	18.8%
Air temperature (K)	14.9%
Process temperature (K)	7.7%
Type (L + M)	2.7%

The three mechanical variables concentrate **75% of total importance**, confirming and quantifying what the conditional distributions suggested.

Product quality accounts for only 2.7%, even though the raw failure rate is almost twice as high for lower-grade products (3.92% against 2.09%). The effect is largely **mediated by operating conditions**: lower-grade products are machined under more demanding regimes, already described by torque and speed. Once these variables are known, `Type` contributes only residual information. This result illustrates the distinction between marginal correlation and conditional contribution, and represents a contribution specific to Shapley value analysis.

Process temperature contributes half as much as air temperature (7.7% against 14.9%). Correlated at 0.88, the two variables carry largely redundant information: the model relies preferentially on one and exploits only the residual of the other.

7.2 Functional forms

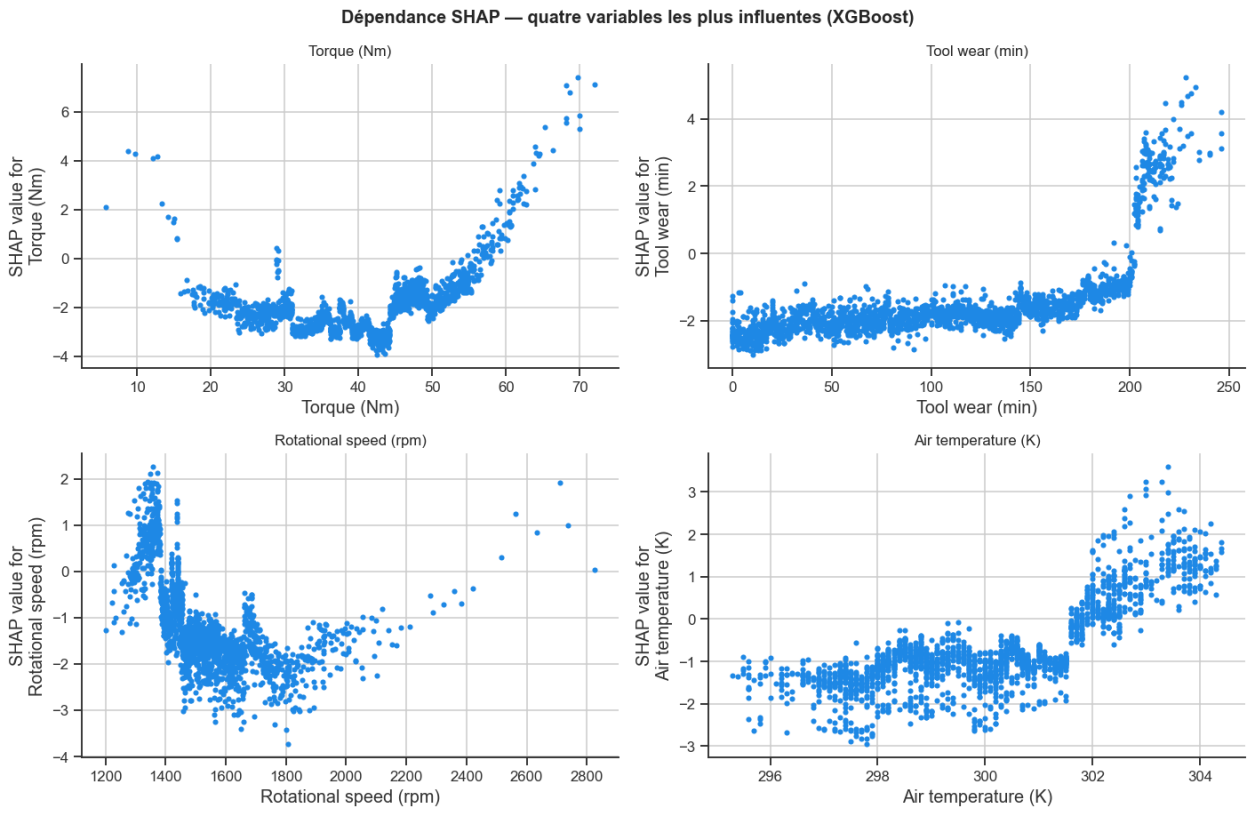


Figure 5. SHAP dependence for the four most influential variables. The horizontal axis shows the variable's value, the vertical axis its contribution to the failure score.

All four variables exhibit distinctly non-linear relationships that logistic regression was structurally unable to represent.

Torque traces a U-shaped curve. Its contribution is minimal around 40–45 Nm — the nominal regime — and rises at both extremes: steeply above 50 Nm, more moderately below 15 Nm. This quantitatively confirms the two failure regimes. A linear boundary, imposing a monotonic effect, must choose one direction and necessarily misses the other.

Tool wear produces a threshold effect. Its contribution remains flat and negative across the entire 0–200 minute range, then shifts abruptly beyond it. The model has thus learned an **implicit replacement threshold**, consistent with the business logic of a tool degrading without consequence until a breaking point.

Rotational speed reproduces the torque U, inverted, the two quantities being mechanically linked at a given power. **Air temperature likewise acts through a threshold**, shifting around 301.5 K.

These four shapes — two U-curves and two threshold effects — explain the performance gap between model families in a single figure.

8. Error analysis

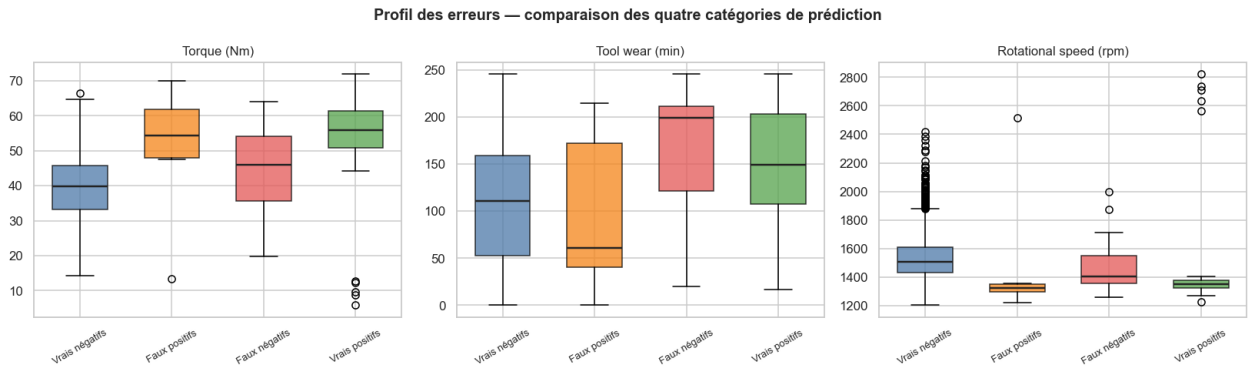


Figure 6. Dispersion of operating variables across the four prediction categories, for XGBoost at the tuned threshold.

Variable	True negatives	False positives	False negatives	True positives
Torque (Nm)	39.5	51.9	44.7	52.4
Tool wear (min)	107.5	94.4	170.3	146.3
Rotational speed (rpm)	1,539	1,430	1,476	1,481
Air temperature (K)	300.0	301.6	300.9	300.7

8.1 False negatives are wear-driven failures

Mean tool wear among false negatives (170 minutes) is the highest of the four categories — exceeding even that of correctly detected failures (146 minutes). Their torque, by contrast, remains moderate (44.7 Nm against 52.4 for true positives).

Examination of individual cases is unambiguous: wear values of 198, 200, 203, 207, 221 and 222 minutes paired with torques ranging from 19.8 to 41.9 Nm, that is, entirely ordinary mechanical conditions.

These observations correspond to tool-wear failures occurring under normal operating regimes after prolonged use. **The model misses them because it relies primarily on instantaneous mechanical stress** — torque accounts for 31% of its importance — whereas these failures arise from a cumulative process.

8.2 False positives are degraded states that did not escalate

The profile of false positives is that of a failure: the highest torque of the four categories (51.9 Nm, exceeding even true positives), the lowest rotational speed, the highest air temperature. These are operating points under mechanical overload that simply did not produce a breakdown.

Such cases are therefore not errors in the strict sense: **they are physically indistinguishable from failures on the basis of the available variables**. From a business standpoint, an alert on these points constitutes early detection rather than a false positive.

8.3 On the decision threshold

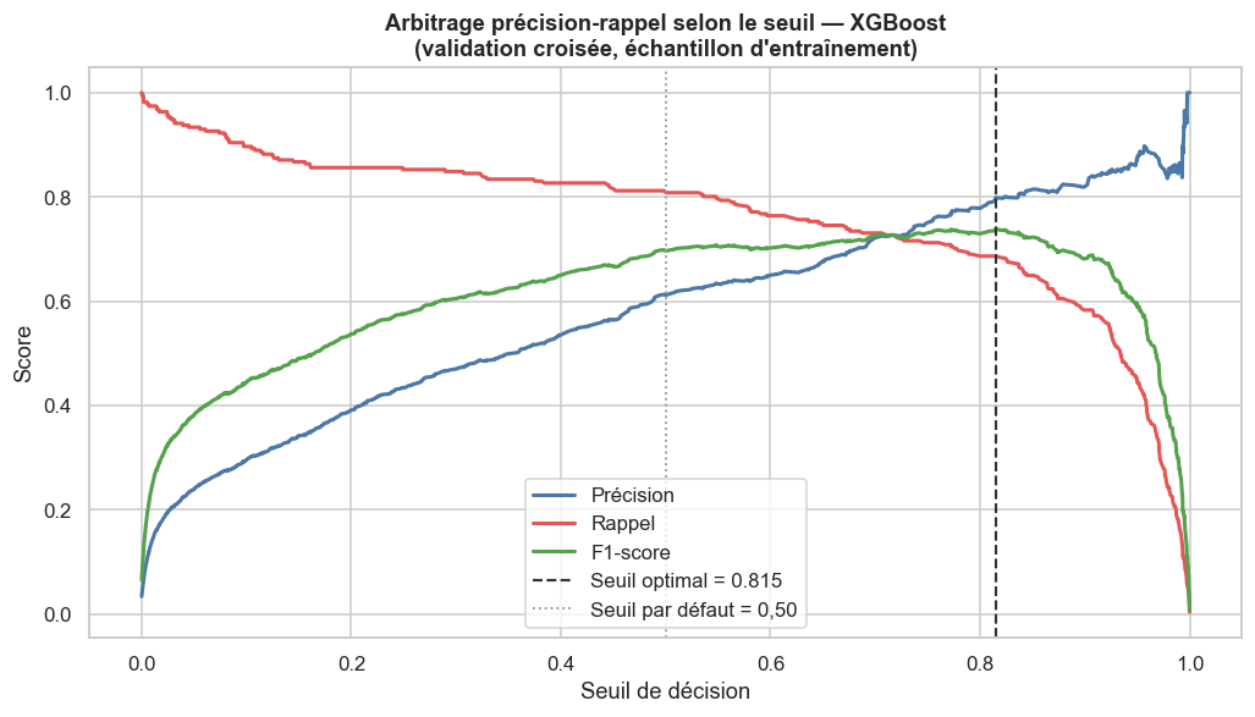


Figure 7. Precision, recall and F1 as a function of the decision threshold, estimated by cross-validation on the training sample.

The optimal threshold identified (0.815) substantially exceeds the conventional 0.50. This shift follows directly from class weighting: by multiplying the weight of failures in the loss function by 28.5, the procedure mechanically displaces predicted probabilities upward.

Tuning raises F1 from 0.700 to 0.762 and cuts false alerts by more than two thirds, but **eight additional failures escape detection**.

F1 weights precision and recall equally, which rests on no business justification. In a real industrial setting, the threshold should derive from the ratio between the cost of unplanned downtime and that of an unnecessary preventive intervention. Should a failure cost ten times more than a false alert, the 0.50 threshold — more favourable to recall — remains preferable. **The threshold is a business parameter, not a statistical constant.**

9. Limitations

9.1 Data-related limitations

Synthetic nature. Relationships between variables are generated by deterministic rules, more regular than would be observed on real sensors subject to measurement noise, calibration drift and missing values. External validity remains to be established.

Labelling defects. Nine failures out of 339 are associated with no identified mode, and eighteen observations carry an `RNF` mode — a stochastic failure, unpredictable by construction — without a failure label. These cases lower the attainable performance ceiling.

Positive class size. The test sample contains only 68 failures. Each misclassified observation shifts recall by 1.5 points: confidence intervals around the metrics are wide, and small gaps between models warrant caution. This applies precisely to the gap between random forest and XGBoost.

Independence assumption. Observations are assumed independent. Were they successive readings from a single machine, this assumption would be violated and the random train/test split would introduce temporal leakage.

9.2 Protocol-related limitations

No hyperparameter optimisation. No systematic search was conducted. The values adopted are reasonable but not empirically justified.

No feature engineering. The dataset's failure modes are defined from derived quantities — mechanical power (the product of torque and angular velocity), the gap between the two temperatures. Constructing these explicitly would likely improve detection, particularly of the wear-driven failures currently missed.

Threshold optimisation criterion. F1 was adopted in the absence of an explicit business cost function, despite reflecting no real operational trade-off.

Collinearity. The strong correlations between the two temperatures and between torque and speed distribute SHAP importance arbitrarily across redundant variables. These values measure a contribution within the model and must not be read as a causal relationship.

10. Conclusion

10.1 Answering the research question

On detection effectiveness. XGBoost attains a PR-AUC of 0.813 against 0.034 for a random classifier — discriminative power twenty-four times above chance. At the tuned threshold it detects 48 failures out of 68 while issuing only 10 false alerts. Automated failure risk detection therefore appears **operationally viable** on this dataset, provided the threshold is calibrated on the business trade-off rather than on statistical convention.

On the determining factors. Three mechanical variables concentrate 75% of importance: torque (30.8%), tool wear (25.1%) and rotational speed (18.8%). The temperatures follow (22.6% combined), while product quality proves marginal (2.7%) once operating conditions are known.

10.2 Model selected

XGBoost is retained: the highest PR-AUC and the better recall at comparable precision once the threshold is tuned. Cross-validation nonetheless places random forest at a statistically equivalent level; the gap observed on the test sample cannot be presented as an established superiority.

10.3 Methodological contribution

Three lessons extend beyond this particular dataset.

The first concerns the **choice of metrics**: under severe class imbalance, ROC-AUC can reach 0.907 for a model issuing one justified alert in seven. PR-AUC alone reflects the true difficulty of the problem.

The second concerns the **identification of data leakage**: the presence of variables recorded after the predicted event is a silent trap, whose detection depends on understanding the business process rather than on statistical analysis.

The third concerns the **fit between model and problem structure**: the superiority of ensemble methods is explained here by the geometry of the failure regions, not by greater learning capacity in the abstract.

10.4 Further work

1. Construct derived variables (mechanical power, temperature gap, wear × torque interaction) and quantify their contribution through an ablation study.
2. Optimise hyperparameters by randomised search with cross-validation on PR-AUC.
3. Replace F1 with an explicit business cost function and derive the optimal threshold from it.
4. Replicate the protocol on real industrial data to test the external validity of the conclusions.

References

Dataset. Matzka, S. (2020). *AI4I 2020 Predictive Maintenance Dataset*. UCI Machine Learning Repository. https://doi.org/10.24432/C5HS5C

Methods. Chen, T. & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. KDD '16. — Lundberg, S. M. & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. NeurIPS 2017. — Saito, T. & Rehmsmeier, M. (2015). *The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets*. PLoS ONE 10(3).

Reproducible code and results. https://github.com/Yvngval/Predictive-Maintenance-Explainable-ML-